


# The Bruker SPR Remote API

# Overview

The Bruker SPR Control Software's remote API allows client frameworks to connect to the Control Software remotely and run its most important operations like starting batch jobs or maintenance operations.

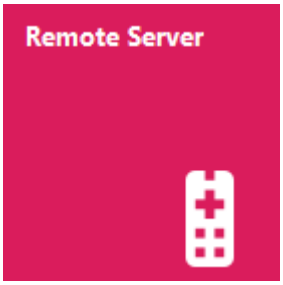
The implementation is based on the Microsoft Windows Communication Foundation (WCF) technology which allows to publish and configure several different endpoints to access the API by just changing the Control Software's configuration file. This allows to write clients using a wide range of different languages, connections, and protocols to talk to the API. In order to keep the API as flexible as possible, it does not support callbacks. Updates about the system's status must be obtained using polling.

As the API is specific for the control software it is highly recommended to have a brief understanding of what you can do with the control software, what the SPR machines of Bruker a capable of, and how they (in principle) work. For that it might be helpful to take a look into the manual of the control software, which can be found in the control software itself, by clicking on "Description" in the right side bar.

# Remote-Control Mode

Before a client can connect to the API, the Control Software must be switched to the remote-control mode. This mode disables most buttons of the user interface so a user in front of the device cannot interfere with the remote client's commands. The user can only check the status of the device and look at the data. The control software opens a server to which a client can connect. The current user is logged of from the control software when the server is started since the remote-control mode uses a dedicated user account with special permissions.

The remote-control mode is started by clicking the "Remote Server" tile on the Control Software's home screen.



You will get to a new page where you can see a graph with the application's recorded data and a log table showing messages that are exchanged between server and client. Once the server is started, all navigation buttons of the user interface are blocked, and it is not possible to leave the page while the server is running.

Message boxes that are displayed during the regular operation of the Control Software are intercepted and handled by the server to avoid that the software gets blocked while a dialog is shown. Dialogs that allow more than one possible answer can be configured in the file "RemoteServiceDialogResults.xml" which is part of the Control Software's installation.

# Simulation Mode

The purpose of the simulation mode is to allow users to plan their assays without using a real machine. It may also be useful for the developers of lab automation frameworks who integrate the SPR device into their framework. The software is automatically started in simulation mode if no machine is connected.

There is also a feature to simulate errors. It is implemented as batch command ("Throw Error" which you will find in the Method Editor) which can also be used for nonremote testing. The command has no parameters to select. The command will throw an exception when it is executed during a batch job and cause the job to crash with an error message. In order to use the command, you first have to enable it. To do so you have to go to the folder <installation directory of the control software>/Racks and edit the file `CommandPrototypesSPR32.xml`. At the end of this file you fill find a `BatchMethodElement` with the `<Name>Throw Error</Name>`. Remove `<!--` before `<BatchMethodElement ...>` and the `-->` after `</BatchMethodElement>` and save the file. Afterwards you have to restart the software. To simulate an error, you have to create a new method with the Method Editor and add the command "Throw Error" to this newly created method. You can also add additional commands before and after the "Throw Error" command and you can add multiple "Throw error" commands.

# Endpoint Configuration

The endpoints to which clients can connect are configured in the Control Software's `app.config` file. Endpoints can be configured for different transfer protocols (TCP or HTTP). Clients can either run locally on the same PC like the Control Software or access the API via a network connection. The base

addresses for the endpoints can also be configured. By default, the base address for HTTP endpoints is 9001, resp. 9002 for TCP connections.

Once the remote-control server of the Control Software has been started you can check that it is running by entering the URI `http://localhost:9001/bruker-spr/` into a web browser's address field on the computer on which the Control Software is running. You can also do this from a different computer if "localhost" is replaced with the IP address of the Control Software's PC. Of course, the port 9001 must be open to the outside to do this.

You can also use the JSON endpoint to transfer commands directly from the browser to the control software. Some examples:

- `http://localhost:9001/bruker-spr/GetOperationMode` gets the current operation mode as an integer. The meaning of the value can be picked from the table in the Interface Description section.
- `http://localhost:9001/bruker-spr/GetNamesOfMaintenanceProcedures` will return a list of names of all maintenance commands that can be run remotely in JSON format.
- `http://localhost:9001/bruker-spr/json/RunMaintenanceProcedure?procedureName=Reset%20System` starts the maintenance command to reset the device's stepper motors and pumps. This only works if the Control Software is either idle (resp. paused or completed) or in "Reset Required" state (usually after an error).

# Client Implementation

Because the server provides several endpoints, there are many ways to implement a client. We do not provide any client applications which are intended for real usage, however there are several demo clients that show how a client application can use the API. You can download the code of these demo clients at https://github.com/BrukerSPR/Bruker-SPR-Remote-API.

## C#

The C# implementation can be found in the RemoteAppTestClient project. It is a Windows Forms application where the client object is based on the `System.ServiceModel.ClientBase` class using a `IClientChannel` that implements the same data contract interface as the server. The application provides controls to test almost all methods of the API and connects to the service using webHttp or TCP. The connection can be configured in the application's config file.

## C++

Two options are given to create a C++ client. Option number 1 depends on Microsoft and its `WebServices.dll`. This way is described in the `ReadMe.txt` of the RemoteAppTestClientConsoleNetTCP project. The 2nd option is platform independent, uses gSOAP and is described in the `ReadMe.txt` of the RemoteAppTestClientConsole project. Both options do not have a hard-coded connection and their classes (`*.c`/`*.cpp` and `*.h` files) are generated with the described tools (see the specific `ReadMe.txt`).

Both demo applications are simple console applications providing the same features. Their usage is self-explanatory. After launching them they ask the user to enter a connection URI (or choose the default) and then offer different options which the user can select by entering the corresponding key.

## Python, Java Script or other Web Languages

The webHttp endpoint can also be accessed using an arbitrary web browser. The data exchanged is based on JSON and requests can be submitted by just typing into the browser's address bar. It is hence easy to write a client in "web languages" like Python or Java Script.

A python example can be found in the RemoteAppTestClientPython project. It contains the class `Mass1Api` which allows to call the remote API functions and a `Tools` class providing some helper methods. These two classes make it trivial to write a simple script to enqueue operations. An example for such a script is `RemoteAppTestClientPython.py`.

# Interface Description

In order to keep the client implementation as flexible as possible, our server does not use callbacks. This means that the server is not capable of contacting the client framework actively if an error has occurred or the state of the software has changed. The client must ask in regular intervals instead (polling). Messages by the control software are queued in a thread-safe queue instead. The arguments and return types are restricted to very fundamental types like numbers, Boolean values, strings, and arrays.

# Methods

All available API methods are described in the following. Whenever "method" (or "methods") is part of the name of the API method or its description, not the

API method is meant but the methods that can be created and run with the control software. These methods may also be named procedures.

## AbortScript()

Aborts the currently running script immediately. That can either be a runset or a maintenance command. Returns 'false' if this is not possible because there is nothing running or the running script is already a cancel script (since the device runs a script to clean up the flow cell when interrupted in the middle of an injection).

```
bool AbortScript()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the script was aborted; otherwise, `false`. |

## CloseSideDoor()

Closes the side door after a plate mover has exchanged a plate on the plate deck. This is only implemented for SPR #64 and will throw a NotSupported Exception on MASS-1 platform.

```
bool CloseSideDoor()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the side door was closed; otherwise, `false`. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not SPR #64. |

## CreateRunset(string[])

Creates a runset that contains the methods specified by the passed array of method names and selects the runset for running. Returns 'false' if the runset couldn't be created. Check for error messages in that case. Possible reasons are that the operation mode doesn't allow to select a runset or that one or more methods with the provided names do not exist in the runset library.

```
bool CreateRunset(string[] methodNames)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| `methodNames` | string[] | Ordered array of method names. |

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the runset was successfully created; otherwise, `false`. |

## GetAssayTypeOfCurrentMethod()

Returns the assay type of the method that is currently running or selected for running. Returns an empty string if there is no method selected.

```
string GetAssayTypeOfCurrentMethod()
```

### Returns

| Type | Description |
| --- | --- |
| string | Assay type, or an empty string if no method is selected. |

## GetAssayTypeOfCurrentRunset()

Returns the assay type of the runset that is currently running or selected for running. Returns an empty string if there is no runset selected.

```
string GetAssayTypeOfCurrentRunset()
```

### Returns

| Type | Description |
| --- | --- |
| string | Assay type, or an empty string if no runset is selected. |

## GetAssayTypeOfMethod(string)

Returns the assay type of the method with the given name. Returns an empty string if there is no method with the given name in the runset library.

```
string GetAssayTypeOfMethod(string methodName)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| methodName | string | Name of a method. |

### Returns

| Type | Description |
| --- | --- |
| string | Assay type of the method, or an empty string if not found in the runset library. |

## GetAssayTypeOfRunset(string)

Returns the assay type of the runset with the given name. Returns an empty string if there is no runset with the given name in the runset library.

```
string GetAssayTypeOfRunset(string runsetName)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| runsetName | string | Name of a runset. |

### Returns

| Type | Description |
| --- | --- |
| string | Assay type of the runset, or an empty string if not found in the runset library. |

## GetAssayTypesOfAllMethods()

Returns an array containing the assay types of all methods that are loaded in the current runset library.

```
string[] GetAssayTypesOfAllMethods()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Distinct array of assay types. |

## GetAssayTypesOfAllRunsets()

Returns an array containing the assay types of all runsets that are loaded in the current runset library.

```
string[] GetAssayTypesOfAllRunsets()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Distinct array of assay types. |

## GetCurrentPlateId(string)

Gets the ID of the plate that is currently inside the machine at the provided deck location. Returns an empty string if there is no plate or ID is unknown. This is only implemented for MASS-1 platform and will throw a NotSupported Exception on SPR #64.

```
string GetCurrentPlateId(string deckLocation)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| deckLocation | string | Name of the deck location where the plate is located. |

### Returns

| Type | Description |
| --- | --- |
| string | ID of the plate, or an empty string if no plate is present or the ID is unknown. |

### Exceptions

| Type | Description |
| --- | --- |
| ArgumentException | Thrown if location does not exist on deck. |
| NotSupportedException | Thrown if not MASS-1 platform. |

## GetCurrentSamplePlateId()

Gets the ID of the sample plate that is currently inside the machine. Returns an empty string if there is no plate inside the machine or the ID is unknown. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
string GetCurrentSamplePlateId()
```

### Returns

| Type | Description |
| --- | --- |
| string | ID of the sample plate inside the machine, or an empty string if no plate is inside or the ID is unknown. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## GetDeckLocations()

Returns an array of all deck location names.

```
string[] GetDeckLocations()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Names of all deck locations. |

## GetErrors()

Gets an array of strings with information about errors like 'no data', 'runset error', ...

```
string[] GetErrors()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Array of current errors. |

## GetMessage()

Gets the first message from the server's message queue. A message consists of three strings in an array where the first string is a time stamp formatted like yyyyMMdd-hhmmss.fff, the second gives information about the type of message (Error, Warning, DataSaved, OperationModeChanged, ...), and the third string is the message itself. Call this method on a regular base using polling and check the result for null or use the HasMessage method first. A complete list of error types can be found in Error Types.

```
string[] GetMessage()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Array of three strings representing the message (timestamp, type, message), or `null` if no message is available. |

## GetMethodNamesOfRunset(string)

Returns the names of the methods in the runset with the given name.

```
string[] GetMethodNamesOfRunset(string runsetName)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| `runsetName` | string | Name of a runset. |

### Returns

| Type | Description |
| --- | --- |
| string[] | Ordered array of method names in the runset. |

## GetNameOfCurrentMethod()

Returns the name of the method that is currently running or selected for running. Returns an empty string if there is no method selected.

```
string GetNameOfCurrentMethod()
```

### Returns

| Type | Description |
| --- | --- |
| string | Method name, or an empty string if no method is selected. |

## GetNameOfCurrentRunset()

Returns the name of the runset that is currently running or selected for running. Returns an empty string if there is no runset selected.

```
string GetNameOfCurrentRunset()
```

### Returns

| Type | Description |
| --- | --- |
| string | Runset name, or an empty string if no runset is selected. |

## GetNamesOfMaintenanceProcedures()

Returns an array containing the names of all maintenance commands that are available and can be run without user interaction.

```
string[] GetNamesOfMaintenanceProcedures()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Names of all available maintenance procedures. |

## GetNamesOfMethods()

Returns an array of all method names that are loaded in the current runset library.

```
string[] GetNamesOfMethods()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Names of all methods. |

## GetNamesOfMethodsOfAssayType(string)

Returns an array containing the names of all methods of the provided assay type that are loaded in the current runset library.

```
string[] GetNamesOfMethodsOfAssayType(string assayType)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| assayType | string | The assay type (the list of assay types can be obtained using GetAssayTypesOfAllMethods()). |

### Returns

| Type | Description |
| --- | --- |
| string[] | Names of all methods of the provided assay type. |

## GetNamesOfRunsets()

Returns an array of all runset names that are loaded in the current runset library.

```
string[] GetNamesOfRunsets()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Names of all runsets. |

## GetNamesOfRunsetsOfAssayType(string)

Returns an array containing the names of all runsets of the provided assay type that are loaded in the current runset library.

```
string[] GetNamesOfRunsetsOfAssayType(string assayType)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| assayType | string | The assay type (the list of assay types can be obtained using GetAssayTypesOfAllRunsets()). |

### Returns

| Type | Description |
| --- | --- |
| string[] | Names of all runsets of the provided assay type. |

## GetOperationMode()

Gets the current operation mode of the device. The meaning of the return value can be found in the OperationMode documentation.

```
int GetOperationMode()
```

### Returns

| Type | Description |
| --- | --- |
| int | Integer value representing the operation mode. See OperationMode for further details. |

## GetPlateId(int, string)

Gets the ID of the plate at the provided deck location of the method with the given index in the current runset. Returns an empty string if the index is out of range, if there is no runset, no plate, or the plate doesn't have an ID assigned.

```
string GetPlateId(int methodIndex, string deckLocation)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| `methodIndex` | int | Index of the method in the current runset. |
| `deckLocation` | string | Name of the deck location where the plate is located. |

### Returns

| Type | Description |
| --- | --- |
| string | ID of the plate, or an empty string if not available. |

### Exceptions

| Type | Description |
| --- | --- |
| ArgumentException | Thrown if location does not exist on deck. |

## GetSamplePlateId(int)

Gets the ID of the sample plate of the method with the given index in the current runset. Returns an empty string if the index is out of range, if there is no runset, no plate, or the plate doesn't have an ID assigned. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
string GetSamplePlateId(int methodIndex)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| `methodIndex` | int | Index of the method in the current runset. |

### Returns

| Type | Description |
| --- | --- |
| string | ID of the sample plate, or an empty string if not available. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## GetStandbyAfterFinish()

Gets whether the device should go into standby mode after the runset has finished.

```
bool GetStandbyAfterFinish()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the device goes into standby mode after finishing the runset; otherwise, `false`. |

## GetWarnings()

Gets an array of strings with information about warnings like 'temperature unstable', 'runset warning', ...

```
string[] GetWarnings()
```

### Returns

| Type | Description |
| --- | --- |
| string[] | Array of current warnings. |

## HasErrors()

Checks whether there are any errors that are not covered by the operation mode.

```
bool HasErrors()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if there are errors not covered by the operation mode; otherwise, `false`. |

## HasMessage()

Checks whether the server's message loop contains at least one new message.

```
bool HasMessage()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if there is at least one message in the queue; otherwise, `false`. |

## HasWarnings()

Checks whether there are any warnings that are not covered by the operation mode.

```
bool HasWarnings()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if there are warnings not covered by the operation mode; otherwise, `false`. |

## IsChipDocked()

Checks whether a chip is docked. Returns '1' = docked, '0' = undocked, '-1' = unknown.

```
int IsChipDocked()
```

### Returns

| Type | Description |
| --- | --- |
| int | 1 = docked, 0 = undocked, -1 = unknown. |

## IsReagentPlateTrayIn()

Checks whether the reagent plate tray is inside the machine. Returns '1' = inside, '0' = outside, '-1' = unknown. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
int IsReagentPlateTrayIn()
```

### Returns

| Type | Description |
| --- | --- |
| int | 1 = inside, 0 = outside, -1 = unknown. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## IsSamplePlateTrayIn()

Checks whether the sample plate tray is inside the machine. Returns '1' = inside, '0' = outside, '-1' = unknown. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
int IsSamplePlateTrayIn()
```

### Returns

| Type | Description |
| --- | --- |
| int | 1 = inside, 0 = outside, -1 = unknown. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## IsSideDoorOpen()

Checks whether the side door is open. Returns '1' = open, '0' = closed, '-1' = unknown. This is only implemented for SPR #64 and will throw a NotSupported Exception on MASS-1 platform.

```
int IsSideDoorOpen()
```

### Returns

| Type | Description |
| --- | --- |
| int | 1 = open, 0 = closed, -1 = unknown. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not SPR #64. |

## LeaveStandby()

Leaves the standby mode. Returns 'false' if the machine is not in standby mode when this API method is called.

```
bool LeaveStandby()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the machine left standby mode; otherwise, `false`. |

## MoveReagentPlateTrayIn()

Moves the reagent plate tray into the housing. Returns 'false' if this is not possible. This might be the case if there is a runset or maintenance procedure is currently running or if a reset is required. Check for errors in this case. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
bool MoveReagentPlateTrayIn()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the reagent plate tray was moved in; otherwise, `false`. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## MoveReagentPlateTrayOut()

Moves the reagent plate tray out of the device so the plate mover can get or put a plate from it. Returns 'false' if this is not possible. This might be the case if there is a runset or maintenance procedure currently running or if a reset is required. Check for errors in this case. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
bool MoveReagentPlateTrayOut()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the reagent plate tray was moved out; otherwise, `false`. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## MoveSamplePlateTrayIn()

Moves the sample plate tray into the housing. Returns 'false' if this is not possible. This might be the case if there is a runset or a maintenance procedure is currently running or if a reset is required. Check for errors in this case. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
bool MoveSamplePlateTrayIn()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the sample plate tray was moved in; otherwise, `false`. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## MoveSamplePlateTrayOut()

Moves the sample plate tray out of the machine so that the plate mover can get or put a plate from it. Returns 'false' if this is not possible. This might be the case if there is a runset or maintenance procedure currently running or if a reset is required. Check for errors in this case. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
bool MoveSamplePlateTrayOut()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the sample plate tray was moved out; otherwise, `false`. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## OpenSideDoor()

Opens the side door so that a plate mover can exchange a plate on the plate deck. This is only implemented for SPR #64 and will throw a NotSupported Exception on MASS-1 platform.

```
bool OpenSideDoor()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the side door was opened; otherwise, `false`. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not SPR #64. |

## PauseRunsetAfter(int)

Pauses the runset that is currently running. The pauseMode argument sets when the runset is paused (1 = after current command, 2 = after current cycle, 3 = after current method). Returns 'false' if this is not possible because there is currently no runset running.

```
bool PauseRunsetAfter(int pauseMode)
```

### Parameters

| Name | Type | Description |
|---|---|---|
| `pauseMode` | int | 1 = pause after current command, 2 = pause after current cycle, 3 = pause after current method. |

### Returns

| Type | Description |
|---|---|
| bool | `true` if the runset was paused; otherwise, `false`. |

## ResetRunset()

Resets the status of the runset after it was interrupted. Returns 'false' if this is not possible because the runset is currently running or if there is no runset to reset. Running the reset method multiple times on the same runset is possible.

```
bool ResetRunset()
```

### Returns

| Type | Description |
|---|---|
| bool | `true` if the runset was reset; otherwise, `false`. |

## ResumeRunset()

Resumes the paused runset where it was paused. Returns 'false' if this is not possible. Check for errors in this case. Reasons might be that the operation mode is different from 'paused' or the device is not ready or on error.

```
bool ResumeRunset()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the runset was resumed; otherwise, `false`. |

## RunMaintenanceProcedure(string)

Starts the maintenance procedure with the given name. Returns 'false' if this is not possible. Check for errors in this case. Reasons might be that the device is not ready or there is already something running.

```
bool RunMaintenanceProcedure(string procedureName)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| `procedureName` | string | Name of the maintenance procedure to be run. |

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the maintenance procedure was started; otherwise, `false`. |

## SelectMethod(string)

Selects the specified method for running. Returns 'false' if the method could not be selected. Check for error messages in that case. Possible reasons are that the operation mode doesn't allow to select a method or that the method with the provided name doesn't exist in the runset library.

```
bool SelectMethod(string methodName)
```

### Parameters

| Name | Type | Description |
|---|---|---|
| `methodName` | string | Name of a method. |

### Returns

| Type | Description |
|---|---|
| bool | `true` if the method was successfully selected; otherwise, `false`. |

## SelectRunset(string)

Selects the specified runset for running. Returns 'false' if the runset could not be selected. Check for error messages in that case. Possible reasons are that the operation mode doesn't allow to select a runset or that the runset with the provided name doesn't exist in the runset library.

```
bool SelectRunset(string runsetName)
```

### Parameters

| Name | Type | Description |
|---|---|---|
| `runsetName` | string | Name of a runset. |

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the runset was successfully selected; otherwise, `false`. |

## SetPlateId(int, string, string)

Assigns an ID to the plate at the provided deck location of the method with the specified index in the currently selected runset. Returns 'false' if the method index is out of range or the method with the given index doesn't use a sample plate. Check for errors in this case.

```
bool SetPlateId(int methodIndex, string plateId, string deckLocation)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| `methodIndex` | int | Index of the method in the runset for which the plate ID is assigned. |
| `plateId` | string | ID of the plate. |
| `deckLocation` | string | Name of the deck location where the plate is located. |

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the plate ID was successfully assigned; otherwise, `false`. |

### Exceptions

| Type | Description |
| --- | --- |
| ArgumentException | Thrown if location does not exist on deck. |

## SetSamplePlateId(int, string)

Assigns an ID to the exchangeable plate of the method with the specified index in the currently selected runset. Returns 'false' if the index of the method is out of range or if the method with the given index doesn't use a sample plate. Check for errors in this case. This is only implemented for MASS-1 platform and will throw a NotSupportedException on SPR #64.

```
bool SetSamplePlateId(int methodIndex, string plateId)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| `methodIndex` | int | Index of the method in the runset for which the plate ID is assigned. |
| `plateId` | string | ID of the plate. |

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the plate ID was successfully assigned; otherwise, `false`. |

### Exceptions

| Type | Description |
| --- | --- |
| NotSupportedException | Thrown if not MASS-1 platform. |

## SetStandbyAfterFinish(bool)

Sets whether the device should go into standby mode after the runset has finished. Always return true.

```
bool SetStandbyAfterFinish(bool goToStandby)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| goToStandby | bool | Option whether the device should go into standby mode after the runset has finished. |

### Returns

| Type | Description |
| --- | --- |
| bool | Always `true`. |

## StartSelectedRunset()

Starts the currently selected runset. Returns 'false' if this is not possible. Check for errors in this case. Possible reasons might be that the system is not ready, is on error, or there is no selected runset.

```
bool StartSelectedRunset()
```

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the runset was started; otherwise, `false`. |

## StartSelectedRunsetFrom(int)

Starts the selected runset starting from the method with the given index. Returns 'false' if this is not possible. Check for errors in this case. Possible reasons might be that the system is not ready, is on error, the index of the method is out of range, or no runset is selected.

```
bool StartSelectedRunsetFrom(int methodIndex)
```

### Parameters

| Name | Type | Description |
| --- | --- | --- |
| `methodIndex` | int | Index of the method in the runset from which the runset is started. |

### Returns

| Type | Description |
| --- | --- |
| bool | `true` if the runset was started; otherwise, `false`. |

# OperationMode

These modes describe the state of the device.

| Value | Name | Description |
|---|---|---|
| `0` | `Idle` | The device is ready to run a script which can be a runset or a maintenance procedure. |
| `1` | `Paused` | The device is idle but there is a runset selected that was paused during run and might be resumed. The ResetRunset() API method has to be called before a new runset can be started. Running a maintenance command in between is possible, however. |
| `2` | `Completed` | The device is idle and the selected runset has completed. The ResetRunset() API method must be called before a new runset can be started. |
| `10` | `StandBy` | The device is currently in standby mode. You have to leave standby to get into the idle, paused, or completed mode. |
| `11` | `ResetRequired` | This state occurs after a stepper motor had an error or if a script was aborted due to an error or by user interaction. The maintenance method 'Reset System' has to be run with the API method RunMaintenanceProcedure(string) to get into Idle, Paused, Completed mode. |
| `12` | `MaintenanceRequired` | The device is not ready for operation but needs technical maintenance. Currently not used. |
| `13` | `DoorOpen` | The front door of the device is open. It must be closed before the machine can be used. |

| Value | Name | Description |
| --- | --- | --- |
| `20` | `Running` | There is currently a script running. This can be either a runset or a maintenance method. |

# Error Types

These are the errors returned by GetMessage().

| Name | Description |
| --- | --- |
| `AppMessage` | Other message from script [Not yet implemented in test version 3.5.0.6]. |
| `CycleCompleted` | A cycle has completed [Not yet implemented in test version 3.5.0.6]. |
| `CycleStarted` | A cycle was started [Not yet implemented in test version 3.5.0.6]. |
| `DataCleared` | The recorded data were cleared from the application's memory [Not yet implemented in test version 3.5.0.6]. |
| `DataSaved` | The recorded data were saved to file [Not yet implemented in test version 3.5.0.6]. |
| `Error` | A critical error has occurred and current operation cannot proceed. |
| `InjectionCompleted` | An injection has completed [Not yet implemented in test version 3.5.0.6]. |
| `InjectionStarted` | An injection was started [Not yet implemented in test version 3.5.0.6]. |
| `MethodCompleted` | A method has completed [Not yet implemented in test version 3.5.0.6]. |
| `MethodStarted` | A method was started [Not yet implemented in test version 3.5.0.6]. |

| Name | Description |
| --- | --- |
| `RunsetCompleted` | A runset has completed [Not yet implemented in test version 3.5.0.6]. |
| `RunsetStarted` | A runset was started [Not yet implemented in test version 3.5.0.6]. |
| `StateChanged` | The operation mode has changed (see OperationMode). |
| `Warning` | An uncritical error occurred. |